Reg. No. | | | | | | | | | | |

[18AS731]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

UAV SYSTEMS

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1. a. Discuss the essential features, pros and cons of "Fixed Wing" and "Rotary Wing" UAVs. **06**

b. Explain what is meant by the term "Geo-fencing" and discuss its importance in the development of urban UAVs. **08**

2. a. Using a neat sketch label the main components of a fixed wing UAV and describe each of its functions in one or two sentences. **08**

b. Discuss the advantages and disadvantages of using a low wing design for a fixed wing UAV. **06**

3. a. A fixed wing UAV has a GTOW of 1kg. Ignoring the contributions of the fuselage and the empennage, determine the stall speed of the aircraft given the following parameters: **06**

Wing Area = $2m^2$, the C_{lmax} of the wing airfoil is 1.4 which occurs at 15^0 alpha and the aircraft is flying at sea level.

b. Using a flowchart discuss the design methodology process flow for a quadcopter and provide a short description of the major stages of the design process. **08**

OR

4. a. A quadcopter is equipped with a motor-propeller combination which can generate a thrust of 700 gm at 100% throttle. Determine the maximum AUM of a quadcopter if the T/W ratio is 3. **06**

b. Discuss in detail the important parameters which need to be considered while sizing propeller dimensions for a quadcopter. **08**

5. a. Describe how Shape Memory Alloys (SMAs) can be used to modify a small UAV's wingtips in flight. **06**

b. Sketch and label the block diagram of a UAV autopilot and discuss the functions of the major components of the autopilot with a short paragraph. **08**

OR

6.	**a.**	Discuss the error sources which influence the location information received from a GPS device.	**08**
	b.	Write a short note on the different types of UAV communication systems which can be used in urban and semi-urban environments.	**06**
7.	**a.**	Describe the important components of a Mission Planning and Control Station (MPCS) and discuss the need for an MPCS for UAV systems.	**06**
	b.	Write short notes on the important parameters to be considered for a Drone Photogrammetry Mission.	**08**

OR

8.	**a.**	Sketch a mission plan for a wildlife survey mission using a fixed wing UAV and write a short note explaining the important stages of the mission.	**10**
	b.	Discuss the possible failures which may be experienced while using drones for a photoshoot and provide possible mitigation actions for each failure.	**04**